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Data-driven digital twin-based smart tunnel maintenance system

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ABSTRACT

Tunnel facility management (FM) is crucial for ensuring safety, efficiency, and resilience of tunnel infrastructure. Current FM practices, such as reactive and preventive maintenance, have limitations - reactive maintenance can't prevent failures, and preventive maintenance can't predict asset maintenance needs, leading to costly and inefficient processes. In addition, existing computerized tunnel FM systems face various challenges, including the lack of integrated real-time monitoring information, visualization of assets in a three-dimensional (3D) environment, supporting predictive maintenance, and scaling into long infrastructure with complicated spatio-temporal relationships. This study addresses these limitations by proposing a data-driven digital twin (DT)-based framework that supports predictive maintenance to improve tunnel FM processes and enhance resilience. The proposed framework consists of six layers, allowing the integration of data from monitoring system, FM system, and building information modeling (BIM) models. The framework proposes a flexible tunnel data model and classification system that hierarchically divides the tunnel models, ensuring an efficient data connection from the physical twin to the DT. The system was implemented in a tunnel case study that generates maintenance plans and work orders using historical and current condition monitoring data, and the 3D visualization technology suggests maintenance and repair processes, making the FM decision process more effective. The proposed system detected and predicted the twin state based on a data-driven analysis, and the prediction accuracy of the machine learning models was sufficiently high for use in real scenarios to make FM plans in advance and prevent asset failures. The proposed framework is contributing to the infrastructure resilience by enhancing the tunnel system ability to predict the maintenance tasks and prevent failures using data-driven DT technology.

1. Introduction

The increasing population and urbanization of cities, have driven attention to underground infrastructures such as tunnels. An underground tunnel system consists of a variety of structural elements, including shotcrete, concrete linings, drainage, and other functional facilities, such as lighting, fire protection, plumbing, and ventilation. These elements are surrounded by a nonhomogeneous complex geological environment (Sharafat et al., 2021). During operations, tunnel components face various defects like deformation, cracking, leakage, service disruptions, and equipment failures, necessitating inspections and maintenance for optimal service. Developing a resilient system for tunnel maintenance tasks can significantly reduce the risk of expected failures (Dui et al., 2025). Currently, reactive and preventive maintenance approaches are used for the facility management (FM) of tunnels. Reactive maintenance is adopted after the failure of an asset, whereas preventive maintenance is a schedule-based method that enables an FM team to perform maintenance tasks at predefined intervals. However, these maintenance methods have limitations. Reactive maintenance doesn't prevent facility failures, and preventive maintenance can't forecast facility conditions for proactive scheduling of repairs or extending its lifespan. (Hosamo et al., 2023). Hence, these maintenance approaches should be changed to smart maintenance systems to address their shortcomings. A predictive maintenance approach can be adopted to enhance the resilience that uses historical data from the systems to predict the condition of the asset so that maintenance actions can be planned in advance (Gbadamosi et al., 2021).

In addition to these maintenance approaches, computerized systems are used to manage the FM data during FM activities. One such system is the computerized maintenance management system (CMMS), which stores historical, daily operational and maintenance (O&M), work orders, and other FM-related data. However, CMMS has limitations, including difficulties in locating the facility, extracting the required FM

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information, and scaling the system to a large infrastructure or more complex environments, such as tunnels, that have complicated spatio-temporal relationships (Lee et al., 2018). Additionally, the system faces challenges in integrating real-time monitoring information and visualization of assets in a three-dimensional (3D) environment, and lacks predictive maintenance capabilities (Lu et al., 2020b). In recent years, building information modeling (BIM) have been integrated with such systems to design effective tunneling FM systems (Khan et al., 2022). A BIM-based framework was proposed for highway tunnels (Chen et al., 2020) and utility tunnels (Yin et al., 2020) to facilitate information sharing and communication. Moreover, 3D geographic information system (GIS) has been integrated with BIM for underground infrastructure management to provide a smart operational environment (Khan et al., 2021; Lee et al., 2018). However, these approaches simulate the tunnel lifecycle data in a virtual environment that provides static information. Moreover, these studies focused on improving the visualization aspects and did not integrate real-time condition data or support predictive maintenance. Relying on FM decisions using such systems can result in low efficiency, low accuracy, high latency, and one-sided decisions.

The digital twins (DTs) approach can be adopted as a comprehensive smart solution (Vivi et al., 2019). Due to the emergence and benefits, numerous researchers have provided definitions and interpretations (Khan et al., 2025; Lu and Brilakis, 2019). Several review papers summarizes, DT definitions, challenges, and opportunities across industries (Babanagar et al., 2025; Liu et al., 2023; Opoku et al., 2021; Qi et al., 2021; Semeraro et al., 2021). Most definitions share three fundamental components of building DT: (1) physical components, (2) digital components, and (3) connections. Physical components represents the actual objects or process, digital components replicates the objects in 3D environment, and connections facilitates to connect the physical and digital components using hardware and software tools (Tao et al., 2019).

The digital representations of physical components has been categorized into three categories (Babanagar et al., 2025): (1) digital model, (2) digital shadow, and (3) digital twin. The digital model depicts only the digital replica of the physical components, which is achieved through BIM techniques. The digital shadow is the next stage, where the information from the physical twin have a uni-directional flow with the digital model. Digital twin, on the other hand, have a bi-directional flow of information between physical and digital components. For operational stage, a DT simulates physical infrastructure/assets in a virtual environment by integrating and processing dynamic real-time FM information in an integrated environment (Lo et al., 2021). To maximize the benefits of the collected data, DTs develop machine learning (ML) and artificial intelligence (AI) models to predict the future of the infrastructure (Hallaji et al., 2022).

Within this context, Li et al. (2024) proposed a four-level maturity framework for DT: descriptive twin, reflective twin, predictive twin, and prescriptive twin. Cotoarbă et al. (2025) linked these maturality level with the digital representations of the physical components. For instance, descriptive twin represents the physical components in a digital environment, which corresponds to the digital model. Additionally, the reflective twin depicts the current known state of the physical components linked with digital twin through uni-directional flow of information developing a digital shadow of the system. Predictive and prescriptive twins are high-level DTs, not only monitoring the physical components but also predicting the state and optimizing the process/actions. Predictive twin forecast the twin state, while prescriptive twin optimizes the process using the predictions.

For tunnels, achieving a predictive twin requires integration of data from multiple systems, including IoT sensors, databases, BIM models, and historical systems. Due to the elongdated nature of the tunnels, developing a single tunnel model is not efficient since the tunnel state and predictions may vary along their alignment. A partitioned model, dividing the tunnel into different sections based on the data collections, state variation, and complexity of the system can contribute to achieve an accurate DT system. Partitioning the tunnel into manageable sections enables localized monitoring, anomaly detection, and maintenance forecasting at the component level. This allows more granular insights and a detailed breakdown of tunnel conditions, facilitating proactive and targeted maintenance. Combining IoT data, ML-driven predictive analytics, and a partitioned tunnel model leads to a higher DT maturity level and overcomes the shortcomings of prior FM approaches. Researchers have applied DTs to various operational stages of O&M activities, such as defect detection (Omer et al., 2019), asset monitoring (Ozturk, 2021), anomaly detection (Lu et al., 2020b), automatic control (Akanmu et al., 2014), structural diagnosis (Tahmasebinia et al., 2019), CO_2 monitoring (Arsiwala et al., 2023), and decision-making ((Lu et al., 2020a). DTs can increase project efficiency by reducing time and cost, increasing accuracy, and improving communication during FM activities (Zhao et al., 2022). However, applying DTs to tunnel O&M is challenging because of the complex nature of tunnel structures. Tunnel maintenance tasks are more complicated than other infrastructure projects because they involve long underground infrastructures with complex interactions and dependencies within the spatial structure of the tunnel. Underground projects also present additional difficulties such as unpredictable geological conditions and the need for precise O&M (Shoaib Khan et al., 2021; Wu et al., 2022). In addition, integrating and visualizing data from various sources, such as-built tunnels, the environment, maintenance, and IoT sensor monitoring data, can make it difficult to obtain a comprehensive understanding of the tunnel. In particular, when a partitioned model is achieved, integrating mutli-source data with the DT; system is difficult in real-time. Additionally, the data update requires the system to achieve and reflect the updated state and predictions demanding a computationally efficient system.

In a recent study, a knowledge-driven framework was developed to propose a DT solution for tunneling the O&M process (Yu et al., 2021). However, the proposed framework lacks self-learning capabilities that rely on expert knowledge and domain expertise and ignores environmental data, which are essential for tunnel DT, particularly in the O&M stage. Existing studies also lack a partitioned tunnel data model for localized monitoring and a specialized classification system tailored for tunnel FM. Most importantly, tunnels require predictive maintenance strategies that can automatically detect and predict the anomalous conditions of an asset and take timely action, which are not addressed in the above frameworks. On the other hand, in a data-driven DT framework, maintenance actions can be predicted through the analysis of continuous data using advanced data analysis and ML algorithms that can significantly overcome these issues.

To address the issues and achieve a predictive twin for tunnel maintenance, a system is needed that segment the tunnel model into different units, defines a classification system to integrate the data according to the tunnel components, where each tunnel components integrates their own data from multiple systems, and support state predictions using prediction model. This study aims to develop and implement a novel data-driven DT framework that integrates data from multiple system, proposes a unified classification system, that connect physical and virtual tunnel twin data, and uses ML to predict the twin maintenance state to improve the tunnel FM process. The contributions of the proposed research are as follows:

(1) This research enhances resilience in tunnel FM by proposing a data-driven DT-based framework for tunneling FM. The framework collects real-time O&M data of tunnel assets, integrates them with their digital replicas in the DT environment, and uses ML algorithms that learn from the collected data and predict the asset state at the component level, making it easier to perform operational analyses and plan maintenance actions.
(2) This paper proposes a flexible segmented tunnel data model that hierarchically divides tunnel components to efficiently locate and

integrate various physical data into the digital world, allowing the capture of more detailed information about tunnel assets.

(3) The proposed study defines the necessary information for tunnel FM, along with a classification system and design of an effective web-based user interface. This interface comprises six modules that automatically generate work orders and maintenance reports using data analytics and ML capabilities. It also integrates multisource data and effectively connects physical and virtual assets. Additionally, the proposed framework and developed system were applied to a tunnel case supported by multiple scenarios for different tunnel assets.

2. Literature review

2.1. Tunnel FM

During the operational phase, inspection and maintenance activities play vital roles in the functioning of tunnel facilities (Fan and Zhou, 2021). Reactive and preventive maintenance are the commonly adopted strategies (Hallaji et al., 2022). Reactive maintenance, also called corrective maintenance, is an unplanned maintenance activity adopted after damage occurs to an asset to recover its condition and make it functional. Reactive maintenance does not address the root cause of equipment failure. Therefore, the same problems can occur repeatedly, leading to increased downtime and repair costs. Scheduling repairs can also be difficult because unexpected breakdowns can occur at any time. These types of maintenance are avoided for assets that have a significant impact because they lead to high costs and service delays. Hence, preventive maintenance strategies have been adopted to mitigate the effects of reactive maintenance. These are preplanned or scheduled maintenance activities that check the status of an asset at regular intervals and prevent failures (Sun et al., 2021). While preventive maintenance activities are substantially better than corrective maintenance, failures can still unexpectedly occur and assets may be replaced while still in a healthy state, as future conditions are not accurately predicted (Hosamo et al., 2022).

The predictive maintenance strategy employs data-driven analysis to predict the asset state. It uses continuous monitoring data from IoT sensors and past data from FM databases to predict asset conditions (Hallaji et al., 2022). Predictive maintenance enables the detection of anomalous asset behavior in near real-time, facilitating timely decision-making (Fan and Zhou, 2021). Although some researchers have created frameworks for predicting the maintenance requirements of individual machines and MEP components in buildings, these frameworks are only suitable for single-component models (Cheng et al., 2020; Hallaji et al., 2022). Tunnel facilities are complex multicomponent systems, and frameworks designed for single components are often inadequate (Jia et al., 2022). Furthermore, these studies focused solely on predicting the state of an individual component with little attention paid to the classification and visualization of physical entities into various maintenance categories. Consequently, the development of specialized predictive maintenance frameworks tailored to the unique characteristics of tunnel systems is necessary for effective maintenance management.

In addition to the maintenance process, tunnel systems face severe problems because of the use of traditional computerized FM systems. Current tunnel FM systems use FM data from multiple heterogeneous sources with complicated spatiotemporal relationships, mostly managed in multiple-format systems (Chen et al., 2020). For example, if an asset requires maintenance, the historical information or manufacturing data and properties of the asset are retrieved from the CMMS, whereas the location or space information is collected from the space management system (SMS), each with a different data format and structure. In addition, manual queries for retrieving essential information are key problems that cause service delays. One of the main challenges with these systems is their lack of integration with other systems, particularly those that provide real-time monitoring and advanced visualization. This integration is critical for enabling operators and managers to better understand asset conditions and performance. In recent years, BIM has been used to enhance visualization during the tunnel FM stage (Chen et al., 2020). However, these studies typically focused on enhancing the visualization capabilities and did not fully address the challenge of integrating real-time condition data or supporting predictive maintenance. These problems indicate that tunnels require an integrated 3D-based digitalized maintenance management system that integrates real-time status information and predicts the asset state to support predictive maintenance and improve maintenance strategies.

2.2. DT for tunnels FM

The DT is a fast-growing technology in the civil engineering field, in which a physical asset has a digital counterpart that reflects its characteristics and information management (Opoku et al., 2021). DT is considered a promising digital technology that integrates static and dynamic information, connecting physical and digital assets using advanced digital and sensor technologies such as BIM, laser scanners, cameras, and IoTs. In the FM stage, any information required for the O&M activities can be efficiently retrieved from the DT (Jiang et al., 2022). At its optimum, DT offers historical and current information about infrastructure as well as contextual details, such as environmental and geological data, which can have a more significant impact during operational use (Wu et al., 2022).

DTs are created for asset monitoring by collecting real-time data from the physical twin to receive updated information owing to variations in the physical space within the tunnel. In the past, studies have been conducted to monitor underground infrastructure. For example, temperature and humidity in subway stations have been monitored using BIM and IoT systems (Marzouk and Abdelaty, 2014). In another study, a BIM-based preliminary framework was proposed for sustainable utility tunnel O&M. However, these studies only focused on physical space monitoring, ignoring the spatial structure and monitoring of existing assets within the physical space. The decision regarding the monitored data was based on engineering expertise without a detailed data-driven analysis, making the systems highly subjective.

Valuable insights can be gained by analyzing the DT data. For example, the as-built geometric information from DT has been used for bridge defect detection, retrofitting, and inspection that greatly improves the detection and retrofitting processes (Xu and Turkan, 2019). Bridge structural performance has been visualized using BIM and a large amount of IoT sensor data, which has improved effective decision-making capabilities (Boddupalli et al., 2019). In addition, efforts have been made to analyze non-geometric information from the past and present for asset-state detection and prediction. For example, the vibration frequency over time has been used to detect anomalous pump behaviors (Lu et al., 2020b). A data-driven framework has been proposed for the predictive maintenance of MEP components and pumps to improve dynamic maintenance plans (Cheng et al., 2020; Hallaji et al., 2022).

Growing research on DT applications that provide a high-fidelity digital environment shows that smart infrastructure and maintenance strategies can be adopted efficiently in an updated manner. For example, several ML algorithms can be used to predict asset states. Recently, artificial neural networks (ANN) are utilized to predict nonlinear time-series trends using data from databases and monitoring devices (Cheng et al., 2020). An ANN has an advantage over traditional statistical methods for learning the nonlinear features of time-series data, and works efficiently in predicting an asset state. Similarly, support vector machines (SVM) have been used as decision support tools for predicting asset conditions (Fan and Zhou, 2021). The precision of machine learning algorithms depends on the training dataset utilized for model training and improves with an increased number of datasets. In addition, the types of input parameters were selected for each component based on their physical twin characteristics.

Since DT are designed to mirror their physical counterparts, the attributes of the physical twin inevitably influence the architecture and design of DTs. In other words, the features and properties of a physical asset have a direct impact on the structure and function of its corresponding DT. Therefore, it is essential to consider the characteristics of the physical twin while developing a DT to ensure that it accurately represents the real-world asset and can effectively replicate its behavior and performance. Tunnels are complex and elongated infrastructures surrounded by geological entities, and their DT can significantly improve the productivity during FM tasks, particularly during the O&M phase. However, the application of DTs in tunnels is challenging and vastly different from that in buildings, bridges, and roads, which require specialized analysis techniques (Jiang et al., 2022). The spatial structure, data collection, requirements for FM tasks (reactive, preventive, and predictive tasks), and the structure of data for creating a dynamic database in tunnels are unique. Currently, the application of DTs in tunnel O&M, particularly in supporting predictive maintenance strategies, has not been thoroughly examined. Hence, this article proposes a framework that implement a DT for tunnel FM.

3. Research problem and methodology

Current FM systems have several limitations that impede their ability to effectively maintain tunnel facilities. First, maintenance activities are reactive and conducted after facility failures occur, which is uneconomical and time-consuming, leading to increased downtime and reduced productivity. Secondly, routine inspections are conducted, which are planned based on manufacturing or construction data, and decisions are made using theoretical data or expert judgment instead of actual measured data (condition monitoring) from the field. These inspections may not capture the actual condition of the assets, leading to inadequate maintenance and increased risk of asset failure. Third, multiple computerized systems are used for FM data management, which often result in service delays in fulfilling requests, increasing downtime, and decreasing efficiency. Furthermore, data collected during facility inspection or maintenance are stored in fragmented and disconnected systems, making them difficult to access and analyze. Additionally, the collected data alone are insufficient for effective tunnel maintenance planning and execution. Advanced visualization, ML, and data analytics functions, are necessary for condition monitoring, condition assessment, performance prediction, and decision support. Hence, the maintenance strategy must be changed from the traditional to a smart maintenance approach to address these limitations (As-is vs. To-be in Fig. 1).

A data-driven DT-based system that can support 3D visualization, integration of real-time data, asset state prediction, and failure classification, is required to ensure better tunnel maintenance. The system presented in Fig. 1 is expected to enhance maintenance strategies adopting real-time conditions from sensors and other platforms to predict the maintenance state, enabling the detection and prevention of asset failures before they occur. Furthermore, maintenance quality can be improved by providing maintenance teams with 3D visualization, generating digital maintenance plans and work orders, and ensuring that maintenance tasks are consistently performed and at a high standard. A framework is proposed in this study to achieve the system shown in Fig. 1 (To-be system), which is shown in Fig. 2. The DT framework proposed in this paper primarily comprises data acquisition, transmission, modeling, integration, analysis, and application layers. The details of each layer are as follows.

3.1. Data acquisition layer

Data acquisition layer collects and stores the data required for the operational stage of the tunnel. It collects design, construction, and operational phase information, such as day-to-day maintenance and operation data, through various methods, including monitoring and manual inspection, to gather the operational status of the tunnel system. Data are collected, including evaluations of the structural performance and defects, maintenance records of each component, life cycle evaluations, quality control, major and minor repairs, replacements, recommended upgrades, and contracts with contractors and installation companies for day-to-day operation and maintenance activities. The as-built information on civil, structural, and functional systems, was monitored in real-time to provide a reliable, safe, and efficient level of service.

The tunnel FM data for DT are divided into physical and digital data. The physical data include tunnel physical entities, data from installed IoT monitoring devices, and manual inspection and maintenance data, which are available in various formats. Real-time IoT operational data are collected for physical spaces, structural entities, and functional components to ensure safe and effective tunnel services.

The digital data are obtained from the BIM tunnel models and the FM system. BIM models contain physical tunnel geometrical and non-geometrical information that portray the as-built condition. The FM system includes historical tunnel maintenance-related information, such as component age, installation date, installation cost, time of inspection, maintenance per year, and cost of maintenance.

3.2. Data processing/transmission layer

The transmission layer acquires physical and digital data from the data source layer and provides them to the other layers. It transmits physical data received from IoT sensors, such as temperature, humidity, vibration, airflow, velocity, and settlement, as well as manual inspection and maintenance data in various formats, from the physical part of the tunnel to the digital part. In addition, the transmission layer transfers digital data, such as geometrical and non-geometrical tunnel information from BIM models and historical maintenance data from the FM system, to the data modeling layer. This layer plays a crucial part in linking the physical and digital parts of the tunnel DT and requires various communication technologies, including short-range and wide-coverage access networks, to ensure efficient and reliable data transfer. Moreover, the transmission layer can use an application programming interface (API) to enable seamless and secure data exchange between systems, such as historical data from databases or BIM data, to the DT system.

3.3. Digital modeling layer

Digital modeling layer is used to build the virtual counterpart of each physical facility in a digital environment. It leverages BIM capabilities to model virtual assets, supplementing nongeometric information that is significant for O&M operations. Furthermore, it develops a relational database that collects real-time data through a transmission layer that can be integrated with the BIM for asset status visualization. A flexible partitioned 3D-based data model is proposed for tunnel assets, making it easy for stakeholders to efficiently locate, link, retrieve, and update FM information during maintenance activities. A voxel-based representation is used for the tunnel surrounding the geological environment to show nonhomogenous characteristics for comprehensive tunnel DT. The BIM virtual tunnel assets and FM semantic data are visualized in a digital environment to analyze the operational state of the tunnel elements along with the surrounding environment in more detail to make better decisions. The details of creating the 3D environment are provided in the next section.

3.4. Data/model integration

Data/model layer integrates tunnel multisource spatiotemporal data into a single collaborative updated platform. In this context, it integrates BIM data and historical tunnel FM information with current monitoring and O&M information. The data and models in the

As-is	To-be
Processes	
• Corrective maintenance strategies • Actions after failure • Scheduled maintenance • Planning based on manufacturing or construction data • Manual inspection and traditional tools • Decisions are based on expert judgment	• Digital twin-based connecting physical and digital twin • Predictive maintenance strategies • Actions based on data driven analysis • Planning based on real-time operational data • Early detection of potential issues • Decisions are based on engineering analysis
Systems	
• Computerized Multiple format systems • Lacking visualization, historical database, integrating real-time data, and scaling to larger infrastructure • Supervisory control and acquisition (SCADA) • Use of 2D layouts for location tracking • Lack of integration with other systems and automatic update • Lack of prediction capabilities	• Digital twin-based data driven system • Support visualization, real-time monitoring, asset state prediction, and failure classification • Physical and digital twins are connected for assets • Use advanced tools such as machine learning for maintenance prediction • Enhances accuracy, reduce downtime, and increases response time for maintenance requests • Supports tunnel spatio-temporal relationship

Fig. 1. Comparison of the current FM practices (As-is) with the proposed system (To-be).

integration layer are integrated, processed, and manipulated, and different functions, such as data analytics and ML, can be used for informed decisions and facility anomaly detection. An appropriate data structure is required to efficiently integrate the FM information with each tunnel segment for accurate O&M decisions. A unique identifier plays a key role in the data-integration layer.

3.5. Analytical layer

Analytical layer leverages data analytics and ML to make data-driven decisions in the DT environment. Algorithms are designed to identify patterns and anomalies in data that can be used to predict future behavior and identify potential issues before they occur. These predictions can then be used to optimize the O&M process and maintenance schedules, reduce maintenance costs and time, and improve the productivity of the FM staff.

3.6. Application layer

The application layer is the implementation layer that enables connections between the stakeholders and the tunnel DT system. It provides different services to facility managers, which serve as a basis for decision-making. The proposed DT system supports tunnel maintenance activities such as reactive, preventive, and predictive maintenance. In reactive maintenance, after a failure has occurred in an asset, the DT service creates an alarm by visualizing the asset and automatically creates a work order retrieving the maintenance information and other historical working conditions of the asset, including installation date, manufacturer information, reason for failure, materials, and properties. These details can help FM managers to repair and maintain faulty components. Furthermore, it keeps the system updated by incorporating the details of the faulty component into a database that can be used for future predictions.

Managers use DT services in preventive maintenance to determine the date, location, and maintenance assets. The managers click on the maintenance asset and list all the required details. This service further helps assess the health status of each planned segment by integrating real-time information through condition monitoring. Most importantly, the maintenance assets and surrounding environment represented by the as-built voxels are visualized in the DT system for a better understanding of the task. The predictive maintenance services are used to predict the future state of an asset. It predicts the maintenance state of a facility using ML and data analytics based on the collected data. In addition, it detects anomalous behavior of an asset from real-time IoT data.

4. Developing DT-based tunnel FM system

The DT-based tunnel FM process is illustrated in Fig. 3. DT is based on a web-based tunnel FM system that links the digital and physical twins for FM decision-making. It connects the digital and physical twins realizing information exchange among the BIM virtual models, IoT-based monitoring system, and FM database. In addition, the system provides a user-friendly digital platform that supports visualization, information integration, and analysis, generates task alerts and work orders, assigns FM tasks and resources, and predicts the future state of an asset.

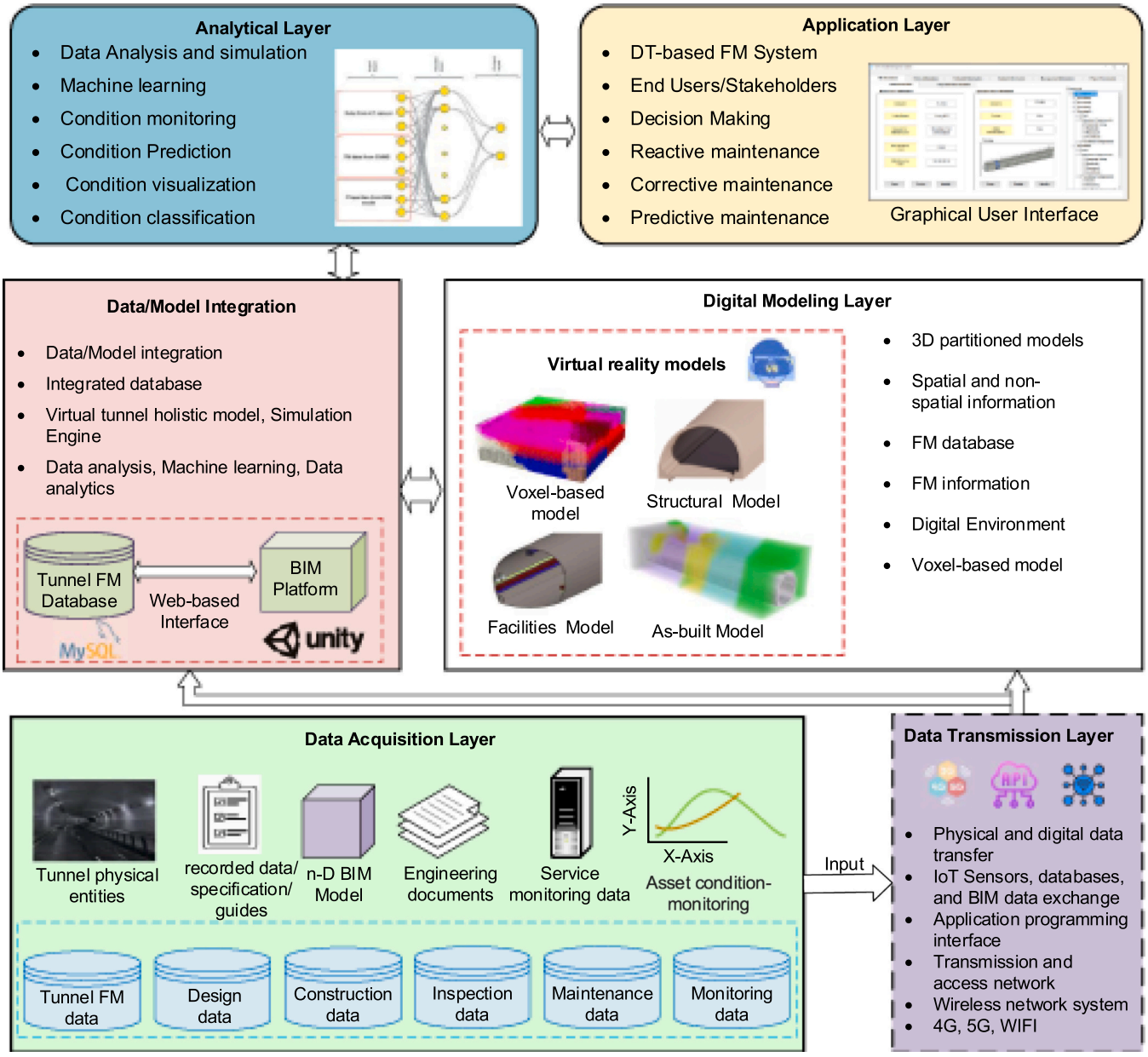


Fig. 2. Proposed data-driven DT-based framework for tunnels FM.

The DT framework first develops BIM virtual models represented by a partitioned model (explained in Section 4.1). Each partitioned segment is linked to the twin information that is updated continuously. Subsequently, a DT system is developed that contains historical FM information along with information from the BIM model and real-time data from the monitoring system (explained in Section 4.2, and 4.4). Real-time monitoring data are obtained from physical spaces as well as from individual critical components. Information from BIM, historical FM, and monitoring systems is integrated with the developed database. To address the issue of integrating and synchronizing information from multiple systems, a unique identifier (ID) is used. A classification system specialized for the unique characteristics of the tunnel system was proposed in this study for efficient information exchange among the systems (detailed in Section 4.3). The classification system is developed based on the partitioned model, where each tunnel segment/component and their corresponding data from the three systems share same ID. The proposed DT system uses data collected from various sources and applies ML algorithms to predict the asset state (detailed in Section 4.5). The model and information are visualized in a DT environment, which makes it easy for managers to make FM decisions (see Section 4.4). Finally, a web-based user interface based on a typical client/server (C/S) was designed with advanced functions to support FM tasks and provide updated real-time information to end users. This interface retrieves data from the central database utilizing IDs, guaranteeing that modifications in BIM, FM, or IoT sources are consistently and very instantaneously represented across all views.

4.1. BIM models

BIM provides a virtual environment that plays a significant role in the tunnel DT. In this context, BIM plays a twofold role. First, it serves as the input data for the DT. Second, it functions as a graphical representation of the results obtained from the aforementioned procedure, thereby providing visualization. For optimal performance, the proposed framework requires access to a comprehensive tunnel BIM database containing all the relevant information. Creating a holistic tunnel model, including ground, structural, and facility models, is needed for a comprehensive as-built representation. In the as-built condition, graphical modeling, representation, LODs, and the

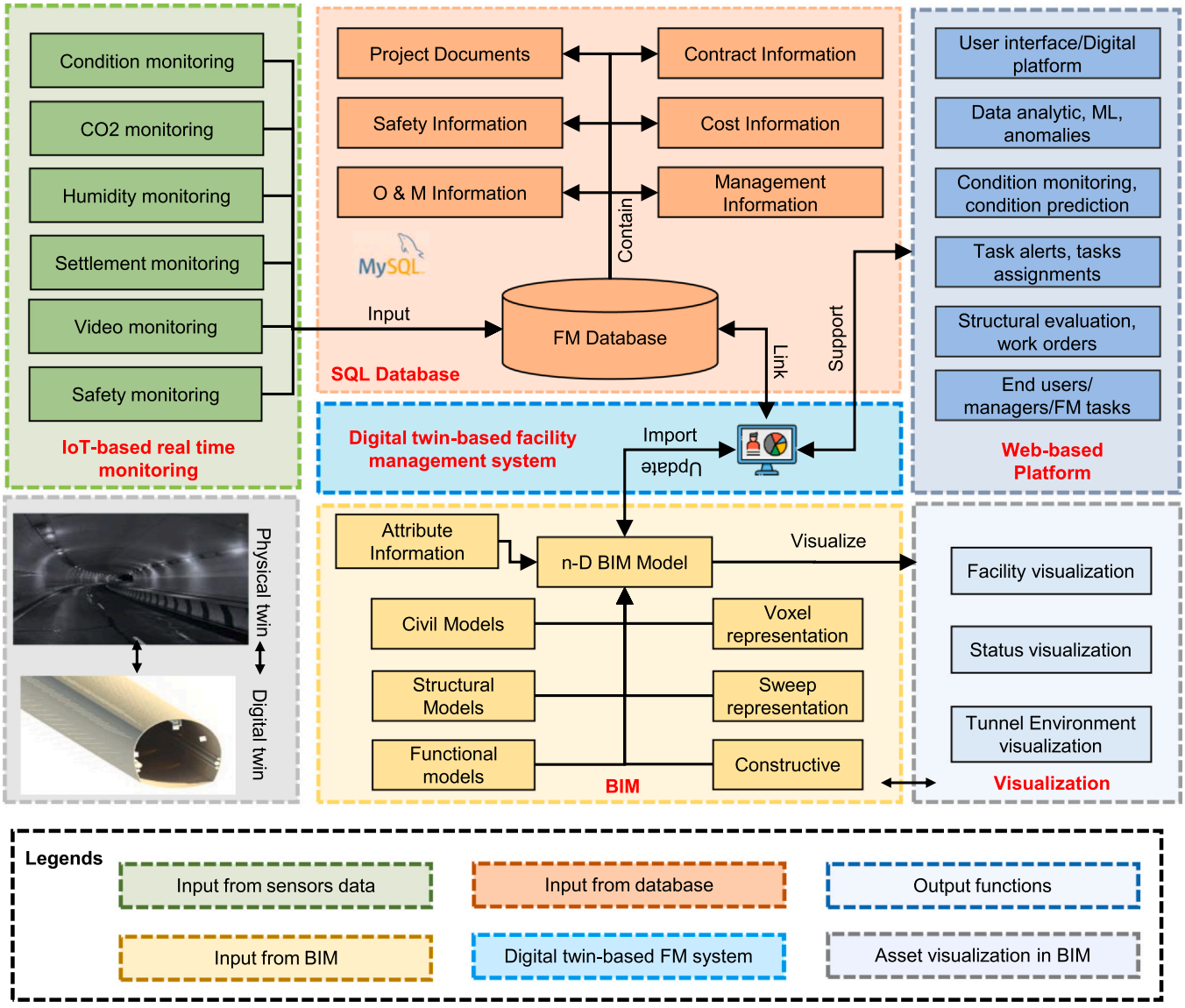


Fig. 3. Process of DT-based tunnel FM representing functions of each layer and data flow.

information contained are important for accurate tunnel DT. A dynamic, spatially enabled segmented tunnel data model was proposed in this study, as shown in Fig. 4a. The benefit of the segmented/partitioned model is that the location-based FM information can be integrated, retrieved, and updated for each segment, leading to an efficient spatial tunnel DT (Fig. 4b). In addition, each segment can be simulated based on a data-driven analysis, allowing the capture of more granular information that leads to accurate FM decision making.

Each BIM element (such as a fan, drainage, or concrete lining) is given a ID in accordance with the partitioned data structure in order to facilitate synchronization with FM and IoT data. This ID acts as the anchor for connecting BIM objects to FM records and real-time sensor feeds. It is maintained as an attribute within the BIM model. Any BIM updates or alterations made during DT operations are recorded with the ID and timestamp, enabling cross-system consistency checks with sensor databases and FM.

Based on the characteristics of the as-built tunnel model and FM requirements, the information and model contained in the holistic BIM model were classified into three categories: civil and site, structural, and functional facilities (shown in yellow in Fig. 3). Civil and site information describes the geographical location of the tunnel, alignment, wear of surfaces, and geological information. Civil information models, such as geological models that represent the tunnel surrounding environment in the as-built condition. The civil information model helps managers visualize the surrounding digital environment for a better understanding of FM site activity.

Tunnel structure information can include concrete linings, shotcrete, tunnel liners, girders, portals, and other support structures. The functional facility models of the tunnel contain mechanical, electrical, ventilation, lighting, alarm, and signal data. These are represented by a hybrid model that includes the swept and CSG representations. The LODs of each BIM component are necessary to provide details regarding the development process, ranging from vague to fully detailed. The LODs of each facility for FM purposes based on their categories are listed in Table 1. A transparent rendering scheme was used for all models except the target category, so that the manager could see the location and surroundings in the holistic model.

A relational database was developed using a Structured Query Language (SQL) for tunnel FM data storage, easy access, and data manipulation. The database stores and manages a significant amount of data on the tunnel FM and provides it to stakeholders during the operational stage. It integrates BIM attribute data, daily maintenance and operational data from FM personnel, historical O&M information, and monitoring data from IoT sensors. The database transfers the stored information to the DT system using an application programming interface (API). A unique ID that corresponds to the ID included in BIM and connected to sensor streams is the first field in

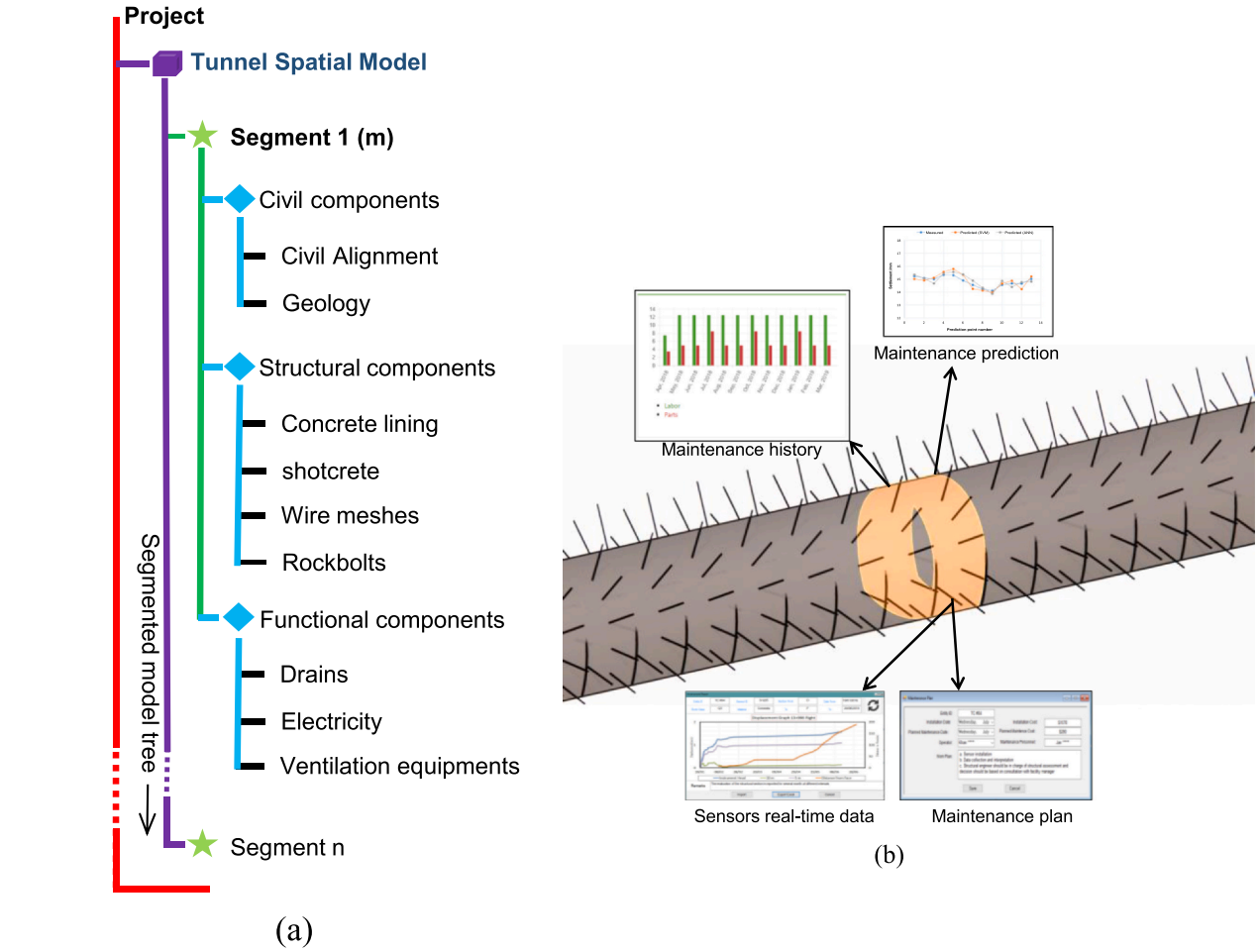
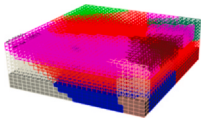
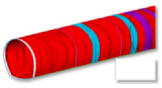
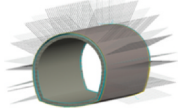
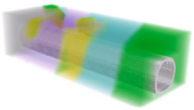
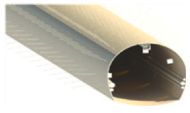
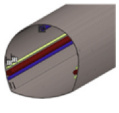


Fig. 4. Proposed BIM data model, (a) The proposed dynamic partitioned-based tunnel data model, and (b) Twin information linked to a tunnel part.

each table to guarantee uniformity across diverse data sources. In all data sources, this ID serves as the primary key. Before being kept, information from sensor readings, BIM attributes, and FM logs (such as maintenance records) is automatically verified against this ID and a timestamp. The system implements pre-established reconciliation criteria or flags data for evaluation if discrepancies or missing values are found. FM data, which is frequently updated on a daily or weekly basis, is occasionally synchronized with semi-static BIM characteristics and high-frequency IoT streams through an ETL (Extract–Transform–Load) procedure. Even when source systems have various naming conventions or update cycles, the system preserves referential integrity between modules by using the ID as the linking key.

Based on the requirements of FM data in tunneling projects, the FM database developed in this study contains six modules: project documents, maintenance and operation information, management

Table 1
Description of BIM models LODs and geometric representation for tunnel FM.

Category	Representation	Models	Details	LODs
• Civil and Site 	Brep, voxels 	Geographical location, geological mapping, alignment, profiles, etc.	ID, location, coordinates, elevation, routes, rock type, RQD, shape, etc.	LOD 500 As-built
• Structural 	Swept + CSG 	Rock bolt, concrete lining structure, shotcrete, lattice girders, and other support structures.	ID, name, shape, material, location, specification, mix design, quantity, etc.	LOD 500 As-built
• Functional 	Swept + CSG 	Pipes, lights, alarms, ventilators, convergence station, etc.	ID, facility type, size, properties, position, location, pipes diameter, materials, etc.	LOD 500 As-built

Database development, data structure, and function requirement

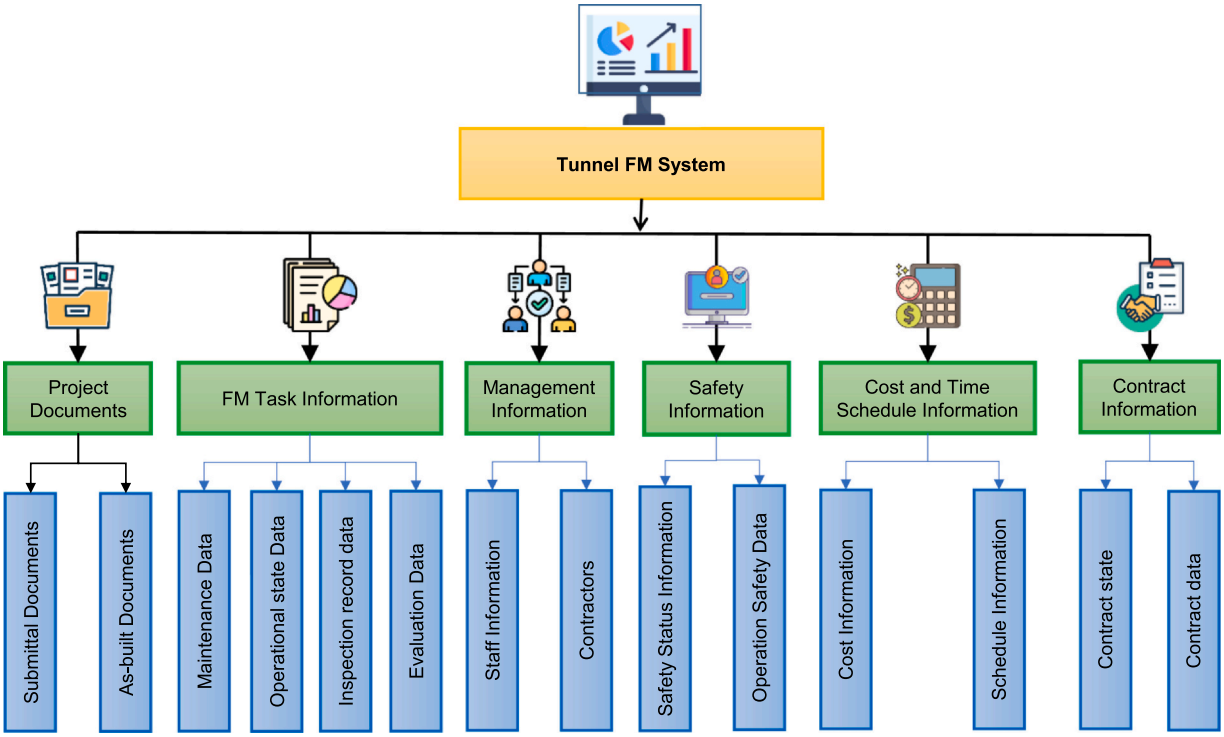


Fig. 5. Tunnel FM system modules and tabs in each module.

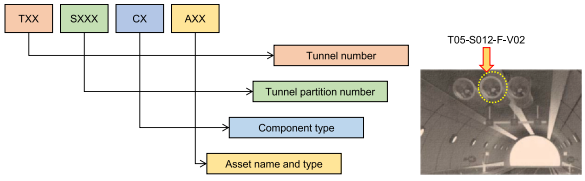


Fig. 6. Classification system for developing unique ID according to the proposed data model.

information, safety information, cost and schedule information, and contract information. The modules are interconnected to complete the tunnel FM database. Furthermore, each module has submodules containing FM information. The structures of the modules, submodules, and data present in each submodule are shown in Fig. 5.

The project document module provides documents that are available after project completion. It includes a library of all files, including O&M and as-built documents. O&M information focuses on the state of the O&M records. In addition, it contains data collected from IoT devices during the inspection and structural evaluation of a facility. Management information includes data on the personnel involved, such as employees and contractors, in the FM task performance because the operation of the tunnel is highly dependent on the quality of work. The information gathered relevant to the safety of the tunnel facilities is stored in the safety management module. The cost and contract information contains financial and legal information about the tunnel facilities.

Each functional module in the tunnel database consists of several tables that store the complete tunnel FM information. The first field of each table in the tunnel database is a unique ID that acts as a primary key to uniquely distinguish data from another and allows users to efficiently move information from the BIM model to the FM system. The proposed system has certain functional requirements for successful development based on tunneling projects. Each tunnel facility is enriched with the information required during the operation stage. Each entity in the tunnel database has a unique identifier for effectively transferring information from BIM and other sources to the FM system. The FM information in the database is constantly renewed to maintain the latest status of the tunnel operation. The information in the tunnel database from different sources is well-integrated, and users can access the information according to their roles and levels of access.

4.2. Tunnel facility classification system

For efficient information exchange between the BIM model, monitoring system, and developed FM database, a relationship must be established. The developed DT system proposes a simplified unique ID based on a partitioned 3D data model. The proposed partitioned data model is based on a dynamic tree model, which forms the foundation for developing unique identifiers for link systems. The proposed unique IDs are combinations of English letters and Arabic numerals. It consists of four levels starting with the tunnel number, partition or segment number, component type (civil, structural, and functional), and asset name and type. Fig. 6 presents the classification method for developing unique IDs and an example of a ventilation fan as the functional component in the 12th segment of No. 05 tunnel. This ID serves as the primary key in FM relational database tables, is connected as metadata in sensor data streams, and is included as a property in BIM models. Cross-referencing between systems with various data structures and update frequencies is made possible by this consistent ID model.

4.3. The integration of real-time condition data and visualization with the DT system

During the operational stage, data are collected in real time from the physical twins (tunnel spaces and components) to ensure a safe and efficient level of service. Real-time data are collected using various sensors. The collection, storage, integration, and visualization of real-time data is an essential step forward for updating tunnel DT and meeting tunnel operation functional requirements. The framework gathers data from various sensors to integrate monitoring information with the DT-based FM system to create a real-time interaction between the physical and virtual environments for informed decisions. The process of integrating the real-time data into the DT system is shown in Fig. 7. This process comprises (i) the collection and storage of the sensor

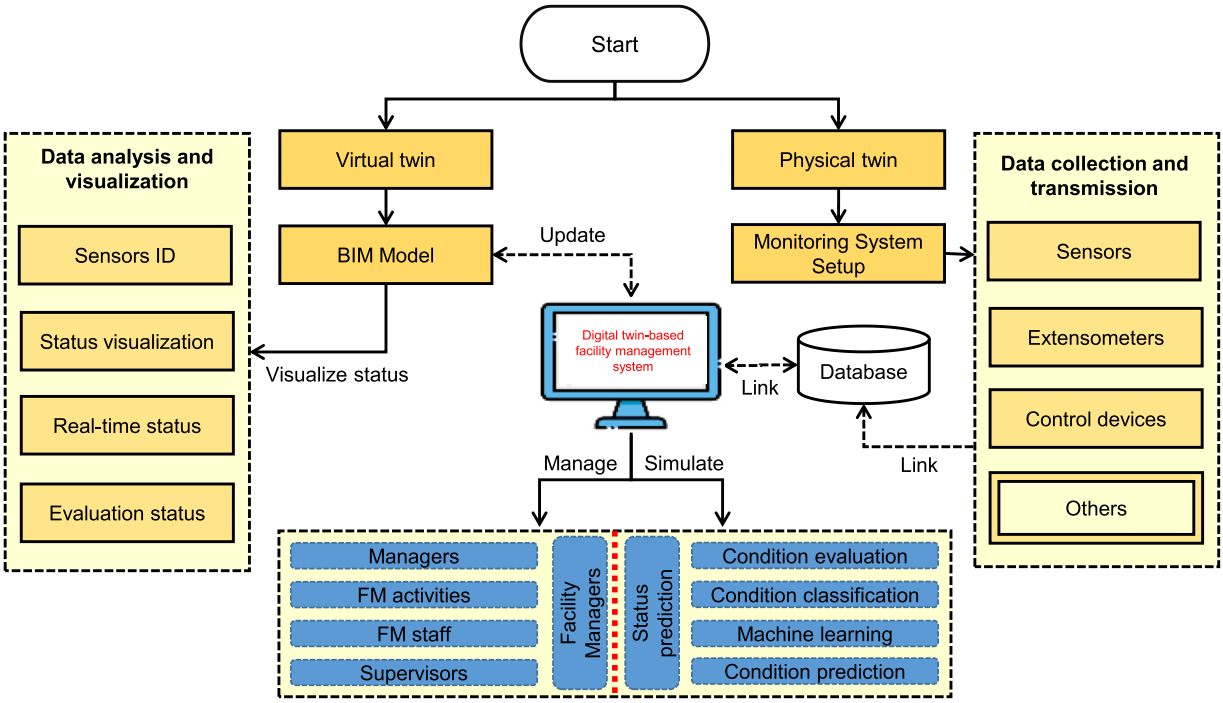


Fig. 7. Process of integrating real-time data with DT system.

data in a relational database, (ii) linking the stored data with the DT-based FM system using an API, (iii) simulation of the collected data, (iv) visualization of the geometries based on the monitored data, and (v) critical decision-making.

Sensors collect data from physical elements in tunnels, such as concrete linings or electromechanical equipment, and integrate them into an FM database that is linked to the DT system. The monitoring data were stored in the developed database and retrieved into the DT system using the open database connectivity (ODBC) API. The DT system simulates the asset conditions using data analytics and ML. The system had predefined rules and a preset threshold interval. When the system collects monitoring data, it simulates the asset condition, calculates the corresponding index, and compares it with a threshold interval indicating the real-time asset condition. The system updates the virtual twin by analyzing the physical twin and visualizes the status of the information in 3D BIM using different colors.

The proposed system classifies the asset status into five categories: excellent, good, fair, poor, and critical. This classification is based on asset health conditions. Each category was visualized using a different color. The information visualization process in a virtual twin can be seen in Fig. 8. Based on the visualization, if the asset condition is normal, the DT system retrieves the information into a database for historical data accumulation. However, if the DT system shows abnormal events, it provides the necessary information to managers with the help of visualization. Managers use information from the database with historical data about that component, monitoring information from the sensor showing the latest status, and the virtual environment to accurately understand the asset condition and formulate a plan for maintenance.

4.4. Condition prediction

The aim of predicting tunnel facility conditions is to provide decision support for maintenance planning to avoid asset failures. The process of condition prediction using ML consists of the following steps (Fig. 9): (a) dataset collection, (b) selection of an ML algorithm, (c) prediction of the asset state, and (d) decision making. The dataset for the proposed DT system is collected from three systems: a database, BIM models, and sensor data. The database provides historical FM information such as past maintenance records, work order information, and inspection data. It captures degredation trends and intervention histories. BIM provides as-built information such as facility type, location, material data, strength data, geometrical data, and construction or install information. These attributes reflects the physical and design-related factors influencing settlement and deterioration. The sensor data include real-time information collected by the system and previous data stored in the system, which directly capture the as-built operational status of the components. Information is collected and analyzed for each planned segment. The segmented model allows prediction of each component individually allowing more granular insights. Each category of the features add values for the prediction model development. For instance, FM data reflects the past deterioration cycles, BIM provides physical and spatial information of assets, and sensor data reflect the operational status.

Every feature vector utilized for ML prediction is indexed by the ID and time to guarantee that the dataset compiled from BIM, FM, and IoT is synchronized and consistent. By doing this, data drift and misalignment are avoided and all predictor variables are guaranteed to correlate to the same physical asset and time period. Before training and prediction, preprocessing scripts make sure that FM logs (low-frequency), IoT readings (high-frequency), and BIM attributes (static) are correctly integrated by ID and arranged chronologically. Additionally, when fresh IoT data becomes available, the model may be incrementally retrained using this ID-based synchronization, preserving historical consistency.

For the predictions, support vector machine (SVM) and artificial neural network (ANN) were selected as the primary algorithms. These two are selected for their appropriateness in the non-linear and multimodal data pattern such as those in tunnels (e.g., settlement data). SVM is effective for small and noisy datasets, which is common in construction datasets. Since, our dataset contains multiple parameters from multi-source (Fig. 9), SVM can also captures nonlinear relationship between input parameters and output parameters using kernel-based mapping. In our study, we used radial basis function (RBF) because of its strong performance in nonlinear regression tasks. The regularization parameter (C) and kernel coefficient (gamma) were tuned through a grid search combined with 5 fold cross validation on the training dataset.

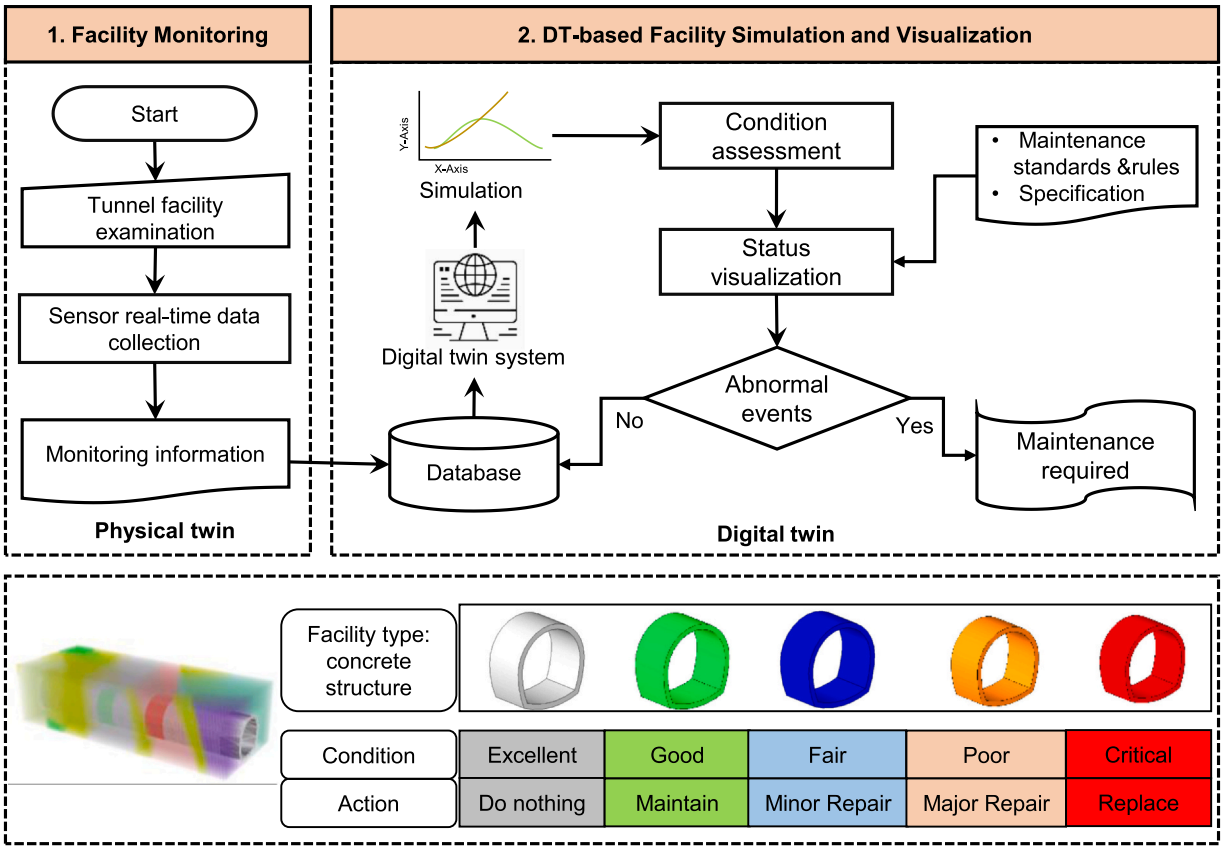


Fig. 8. Information visualization and maintenance classification process in DT-based FM system.

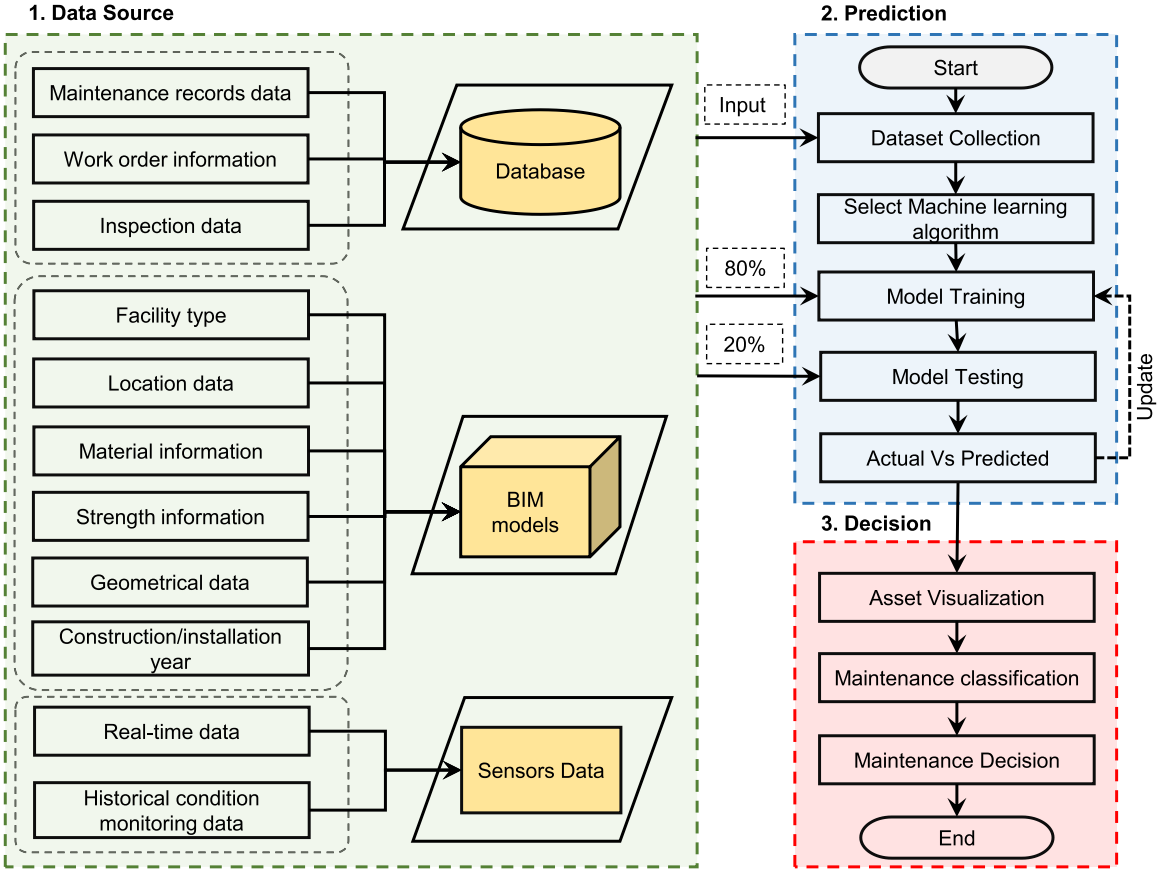


Fig. 9. Asset condition prediction process using machine learning.

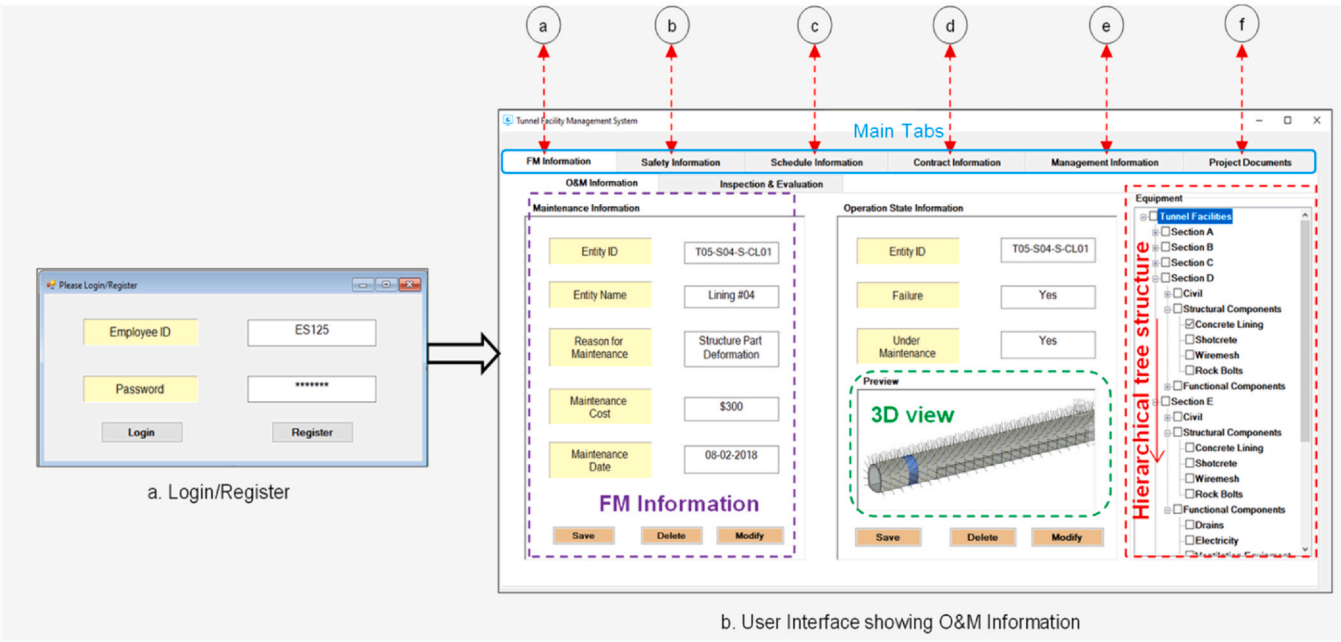


Fig. 10. Graphical user interface (GUI) of the developed DT-based FM system.

The other algorithm, ANN model, designed as a feed-forward multilayer perceptron (MLP) with an input matching the number of features, 3 hidden layers having 16, 8, and 4 neurons, and an output layer having a single neuron. ReLU activation functions were also used in the hidden layers. On the other hand, a linear activation was used in the output layer. Additionally, Adam optimizer and mean square error loss function were used to train. Learning rate and number of epochs hyperparamters were tuned through trail-and-error and cross validation. To mitigate overfitting, early stopping and L2 regularization were applied.

The ML algorithm application selects a dataset and divides it into training and testing data. In this proposal, 80 % of the data were considered for testing and 20 % for training. The training data (80 %) were used to train the algorithm, and the testing data were used to verify the model and predict the asset state. In the training process, 5-fold cross-validation was utilized on the training dataset for hyperparameter optimization and to analyze the generalization performance before the testing. For evaluation, several metrics were used including root mean square error (*RMSE*), mean absolute error (*MAE*), coefficient of determination (R^2), accuracy, and percentage error. In the proposed DT system, the dataset is dynamic and continuous, and the ML model is updated according to new real-time condition data from the sensors.

In the decision-making process, the predicted result is visualized in the virtual model (discussed in the previous section), and the maintenance plan is decided based on the predicted asset condition. The prediction process improves FM practices by predicting the asset maintenance state using data-driven analysis and planning maintenance activities in advance, which reduces the downtime and maintenance costs. By combining IoT-based monitoring, a segmented tunnel data model, and ML-based predictive analytics (SVM and ANN with validated hyperparameters and rigorous performance metrics), the proposed DT framework advances toward the “prediction twin” maturity level and enhances the resilience of tunnel infrastructure.

5. Case study

A tunnel case study was conducted to implement the proposed data-driven DT-based FM system. The case study tunnel was in an operational state and consisted of several pieces of electromechanical equipment intended to eliminate harmful gases. Simultaneously, the tunnel is exposed to settlement that can cause structural damage because it is located on soft ground. The maintenance in the case study tunnel is based on manual scheduling and using traditional FM system that lack 3D visualization and real-time monitoring of the physical assets. In addition, the system has no functions to predict the asset state and enable effective preventive maintenance. This case study was selected to implement the proposed data-driven DT framework, which connects physical and virtual tunnel twins and improves the FM decision-making process.

5.1. DT-based FM system implementation

The implementation of the proposed DT system comprises the following steps: (a) virtual modeling of existing facilities, (b) development of a DT-based FM interface, (c) integration of virtual and physical data with the developed FM system, and (d) scenario implementations.

The first step is to build a virtual twin model that virtualizes all the facilities inside the tunnel and inputs the FM-related attributes and information. Autodesk Revit software was used to develop the virtual twins using the BIM method according to the LODs defined in Table 1. The integration of design and construction information was achieved and is discussed in our previous article (Sharafat et al., 2021). The virtual twin information is integrated with the DT system (explained in Section 5.2). A DT system was developed for the case study tunnel based on the framework described in the Methodology section. The proposed web-based graphical user interface (GUI) for the DT system is shown in Fig. 10.

The GUI of the proposed DT-based FM system included a tree structure for efficient management of the tunnel entities, a 3D view of the BIM model, and information collected through the sensors and FM database. The right part of the GUI presents the tree structure, which demonstrates that the system hierarchically stores and manages tunnel assets. A hierarchical structure was developed for the case study using the proposed DT system (Section 4.1). The first level of the structure divides the tunnel into several parts because it is a geographically long distributed project. The second level of the tree structure includes information on the different structural and functional facilities located and installed in a specific section. The system is sufficiently flexible such that other levels can be achieved according to the conditions, depending on the management purpose. The GUI contains a menu bar

Table 2
Description of the user interface menu bar.

Menu bar options	Contents	Description
a. FM information	O&M, inspection, and evaluation information	The O&M reports can be obtained from this menu. Also, the sensor information from the physical twin can be accessed in the evaluation tab. Furthermore, it has functions to use the historical data from the backend database and the current monitoring data to predict the tunnel status using ML.
b. Safety information	Safety management	Information about safety and standard reports can be found on this page.
c. Schedule information	Cost information, time information	The cost and time information for the operation stage of a selected facility will show in this module.
d. Contract information	Contract information	The contract information and documents for tunnel facilities can be retrieved from this module.
e. Management data	Personnel/employee information	The maintenance staff information and their role is stored on this page
f. Project documents	Engineering documents	Delivery documents and O&M plans can be retrieved from this module.

with six options that provide information stored in each module. Descriptions of the GUI and the six options in the menu bar are given in Table 2. The GUI was specifically created with an intuitive, web-based visual interface to assist non-expert FM managers, hence reducing the necessity for technical competence. The tree structure and 3D BIM visualization enable users to identify and choose assets visually, rather than traversing intricate databases, hence facilitating comprehension of asset locations and interrelations for non-specialists. Every menu option in the GUI offers quick access to essential FM operations, including real-time sensor data, maintenance records, and predictive analytics results, all shown in an accessible way with limited technical terminology.

The graphical user interface incorporates the outputs of the machine learning analytics module directly into the asset view for predictive maintenance. Non-expert users might observe color-coded warnings or alarms inside the 3D model (e.g., emphasizing parts anticipated to necessitate imminent maintenance), accompanied by recommended maintenance activities produced by the system. This enables FM managers to swiftly assess the asset's prospective state without requiring comprehension of the underlying algorithms. Interactive dashboards connected to the GUI visually display trends, forecasts, and anomaly detections, so facilitating proactive decision-making. The GUI visualizes asset conditions over time by superimposing historical and real-time sensor data onto the 3D model. Users may select any asset in the tree or 3D view to access its real-time status, deterioration history, and maintenance data in a streamlined panel. This contextual presentation enables even non-experts to make educated judgments by integrating geographical visualization with data-driven insights.

5.2. Data integration process

The information exchange between the sensors providing real-time monitoring information, a database that stores historical FM information, virtual models, and the proposed DT system, is shown in Fig. 11. The sensor data collected at the case-study location were integrated with the database system using devices such as a gateway that provides interoperability between the sensors from a physical system to a digital database. Sensor information was retrieved from the database using SQL connected via Bluetooth and WIFI or LTE (4 G). Virtual models developed in a BIM environment contain geometric and nongeometric information. The geometric information is integrated with the FM system by an add-in, and the non-geometric information is exported into the database using the ODBC plugin. A database containing sensor information, BIM nongeometric data, and other FM information, including historical data, was linked to the FM system via the ODBC API. This information is matched with each other in the proposed FM system using a unique ID.

5.3. Three scenarios

Considering a case-study tunnel, scenarios are introduced to verify the applicability of the developed FM system. These scenarios include a maintenance plan for scheduled maintenance, work orders for reactive maintenance, and condition monitoring and prediction for predictive maintenance. These scenarios provide examples of how the proposed system supports various FM activities. Different components of the case study tunnel were selected to verify the functionality of the proposed DT system for multiple components. Finally, the benefits of the proposed DT-based FM system were evaluated.

5.3.1. Maintenance plan for scheduled maintenance

The frequencies and descriptions of the FM activities for each planned tunnel asset were incorporated into the FM system from the FM databases according to the tunnel specifications and rules. The tunnel has a longitudinal ventilation system of jet fans, which is reliable for continuous air provision. These fans were monitored to ensure normal operation and detect abnormal problems. The proposed system uses a jet fan as an example to illustrate the system function for the scheduled maintenance of tunnel equipment. Fig. 12 shows the FM plan for the jet fan generated by the system according to the maintenance frequency. In addition, it shows the work order and condition data from the physical fan, including temperature, airflow, and velocity trends that are integrated and provided by the developed DT system. The corresponding virtual twin shows the fan status in green, indicating that the facility needs maintenance because of the increase in temperature. This occurred because of a lubrication problem addressed by the maintenance team. This function helps the manager remind the planned FM activities, monitor the asset condition according to the schedule, perform maintenance activities, and update the facility status into the system. The proposed system provides an advantage in detecting the issue from real-time monitoring information in a jet fan, which is difficult to observe using traditional visual inspection. Additionally, the 3D visualization of the asset makes it easy for FM managers to make maintenance decisions.

5.3.2. Work order for reactive maintenance

The failure of lighting equipment is considered an example of a scenario that has failed during the operational state reported by inspection personnel. The lighting equipment is located by the manager of the proposed DT system. The proposed system issued a work order with the lighting equipment ID, location, manufacturer information, maintenance data, estimated scheduled data, and attribute data, and assigned it to maintenance personnel for replacement (Fig. 13). Following the work order, the lighting equipment was replaced, and the replaced lighting equipment data were fed into the FM system to keep it updated and accumulate historical data. This helps ensure the timely and efficient resolution of lighting equipment failure, ensuring that the tunnel system maintains an effective level of service.

5.3.3. Condition monitoring and prediction for predictive maintenance

The application of the proposed DT system is not limited to scheduled or reactive maintenance, but also supports predictive maintenance that can predict the facility state using condition data from sensors, BIM models, and historical information from databases. The settlement of

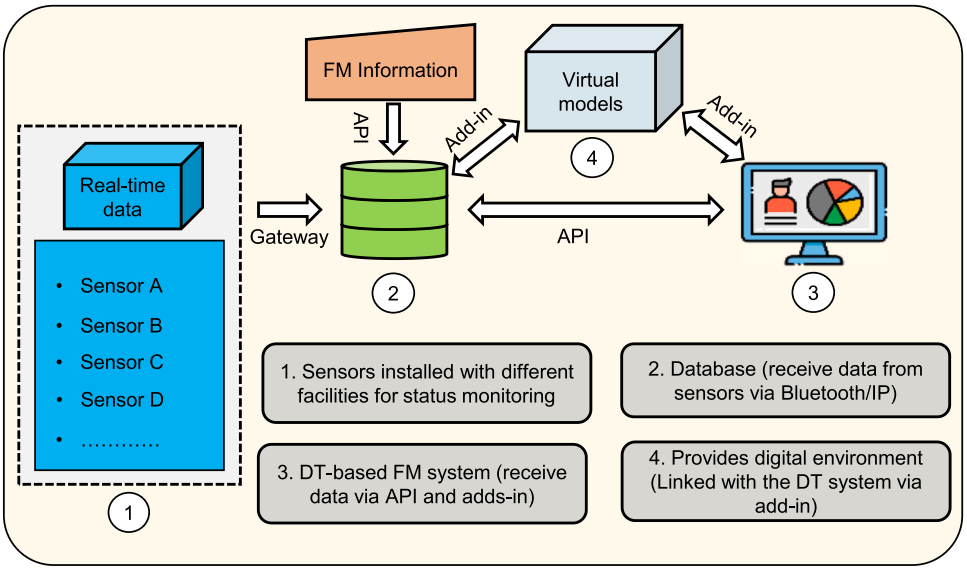


Fig. 11. Data flow among sensors, FM database, virtual models, and FM system.

the tunnel structure was considered as a scenario, which is a key issue in the case study tunnel, to test the applicability of the proposed DT system for predictive maintenance.

Monitoring data are collected from two tunnel segments with different characteristics and environmental conditions. The first dataset was collected from Section H, and the second dataset was monitored in Section D. The surrounding geological conditions and settlement trends are different in both sections. The datasets were collected over 60 consecutive days in June and July. Additionally, it was identified from the measured data that tunnel Section D had been monitored and required minor repairs, which was validated using the prediction feature of the developed system. Furthermore, the section repair was predicted with the application of the ML.

(1) *Condition monitoring*

Fig. 14a shows the locations of the tunnel-monitoring segments surrounded by varying geological conditions represented by the voxels. The settlement monitoring information integrated with the proposed DT system is shown in Figs. 14b and c. The settlement trend in Section H was smaller than that in Section D because the geological conditions were different, as observed in the model, where the geological conditions are differentiated with different colors (Fig. 14a). The proposed system has the advantage of linking the physical and DTs by integrating real-time condition information and leveraging a partitioned-based 3D model that effectively classifies the maintenance action according to the defined criteria. The proposed system successfully integrates real-time

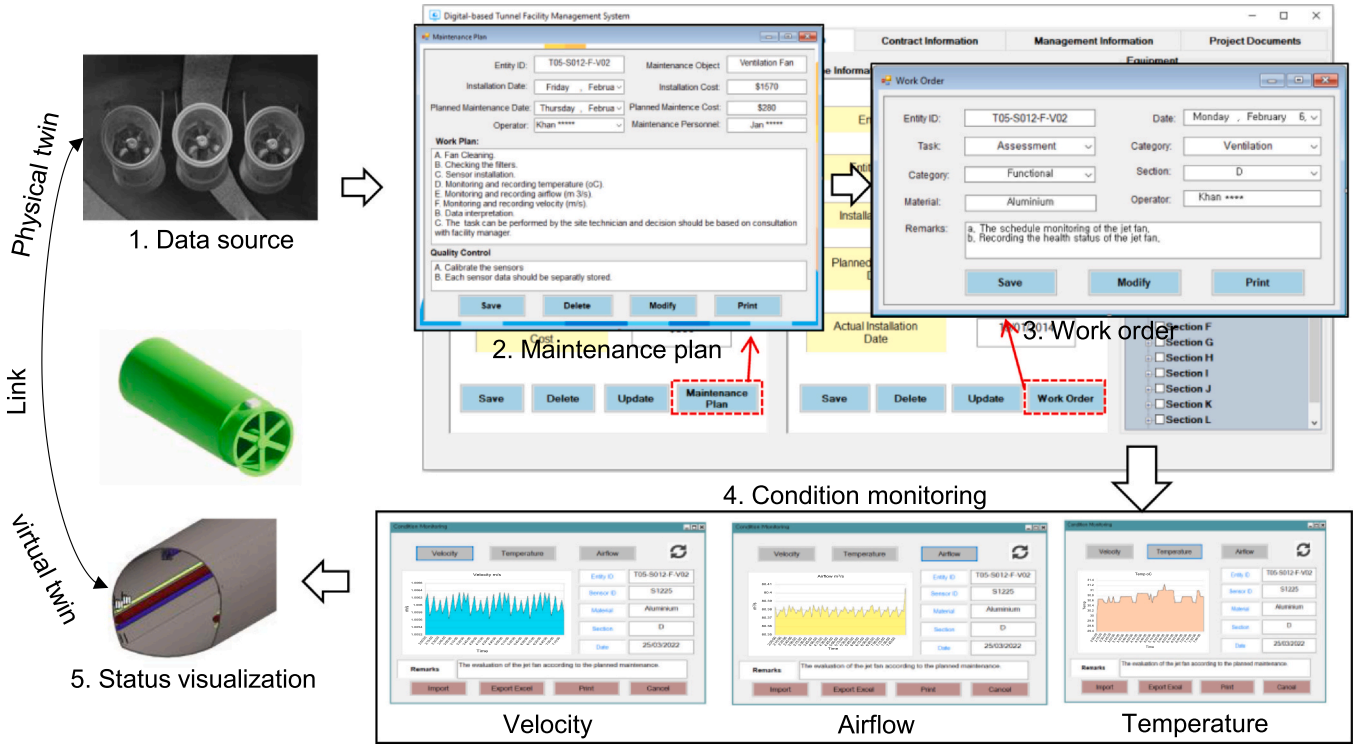


Fig. 12. Schedule maintenance of a jet fan with the proposed system integrating real-time information and status visualization.

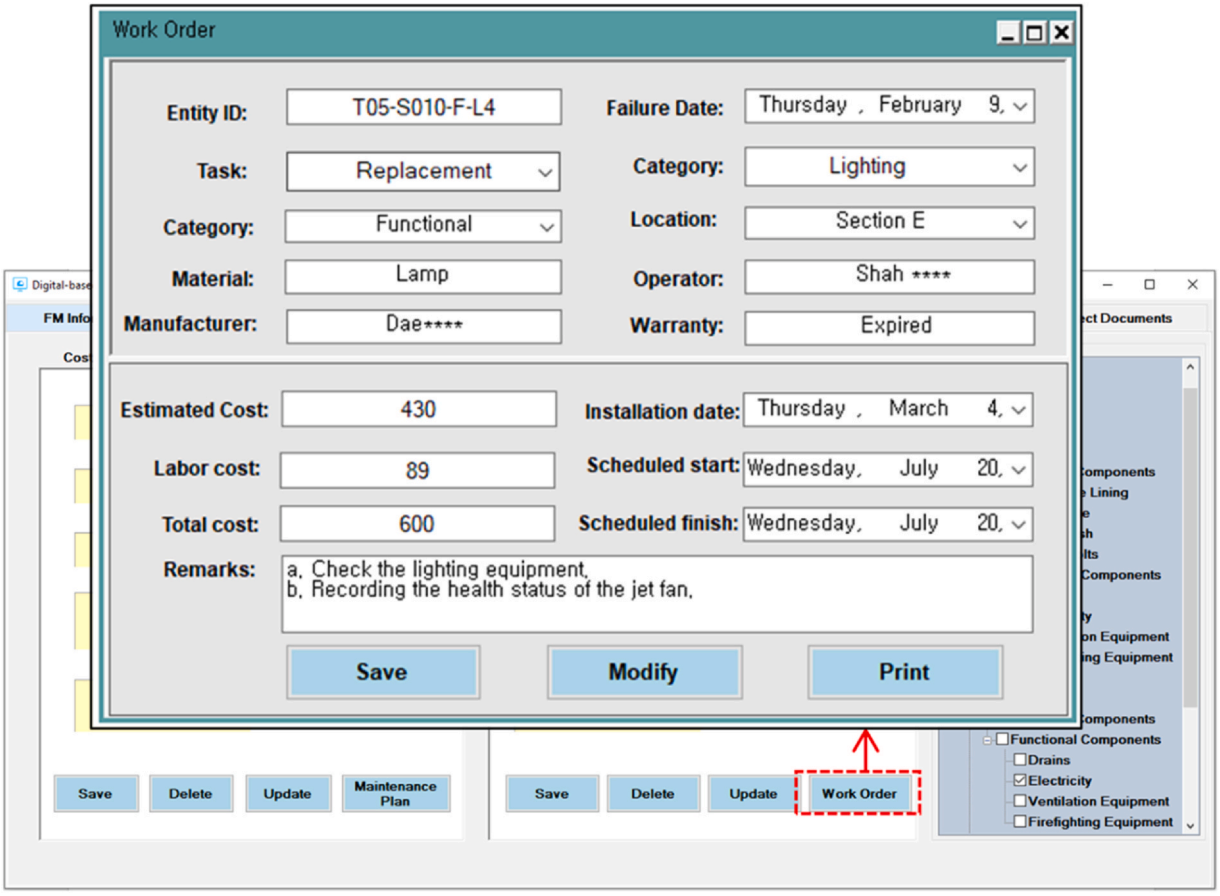


Fig. 13. Work order generation for the corrective maintenance of a lighting equipment.

condition information and detects the abnormal behavior of the tunnel at Section D from the measured data.

The core of condition monitoring was to solve the problem of the lack of real-time data integration with the DT system that could detect issues that are difficult to detect with visual inspection. Tunnel settlement is a serious problem that can occur owing to sudden changes in the tunnel structure during the operational stage or can be caused by the surrounding environment, such as unstable geology. The developed system not only integrated the real-time settlement data, but also detected an abnormal value for Section D, which was visualized by the DT system (Fig. 14c).

(2) *Condition prediction*

Real-time tunnel condition monitoring, data storage, and integration with a DT system are not sufficient for creating a smart maintenance system that can avoid failure. Predicting the future state is necessary to predict abnormal conditions and plan maintenance in advance. However, tunnel settlement is affected by several parameters, such as the surrounding geological conditions, construction, and material of the tunnel structure, and the data pattern is nonlinear, multimodal, and complex. These factors make it particularly challenging to predict tunnel settlement.

The proposed DT system uses ML algorithms, including SVM and ANN, to predict the asset state and improve the FM decision-making process. The process of predicting the condition of an asset is shown in the Fig. 9. SVM and ANN are selected here because SVM is effective for smaller nonlinear datasets with limited noise, while ANN can learn complex, nonlinear relationships in larger feature spaces. For SVM, a RBF kernel was adopted to handle the concrete dataset coming from multisource and their pattern. The regularization parameter (C) and kernel coefficient (gamma) were optimized using grid search combined with 5-fold cross-validation on the training data, ensuring generalization and reducing the risk of overfitting. For ANN, feed-forward MLP was implemented with an input layer corresponding to features from the historical database, BIM, and real-time sensor reading. In this case study, the properties of the tunnel concrete structure from the database, BIM models, and real-time settlement information were used as scenarios to validate the system functionality for predictive maintenance. In addition, geology-type information was used in the dataset input because it significantly affects the tunnel settlement, and each segment has a different geology in the case study location. Along with this, three hidden layers with 16, 8, and 4 neurons utilizing ReLU activation, and a single-neuron output layer with a linear activation function was used for regression. Additionally, the Adam optimizer and mean squared error loss was used in model training. Early stopping and L2 regularization were applied for overfitting avoiding. Trial and error with cross validation was used for learning rate and number of epochs tuning.

The data trends for Sections H and D were different. In total, 60 data points were used for each section, which were collected over two months. Data from June 1–July 17 (80 %) were used as training samples. The remaining 20 % of the data were used as the testing data. 5-fold cross-validation was conducted on the training data to tune hyperparameters and ensure generalization before applying the models to training settlement trends. Importantly, the DT framework is designed to continuously update the models as new IoT sensor data becomes available, ensuring that prediction accuracy improves over time and that the DT remains a living, adaptive system.

The settlement-monitored data and data predicted using the SVM and ANN are shown in Table 3 for section H and D. The settlement prediction results obtained using the proposed DT system for both the segments are presented in the Fig. 15. The blue line represents the measured value collected using the sensors, and the orange and grey

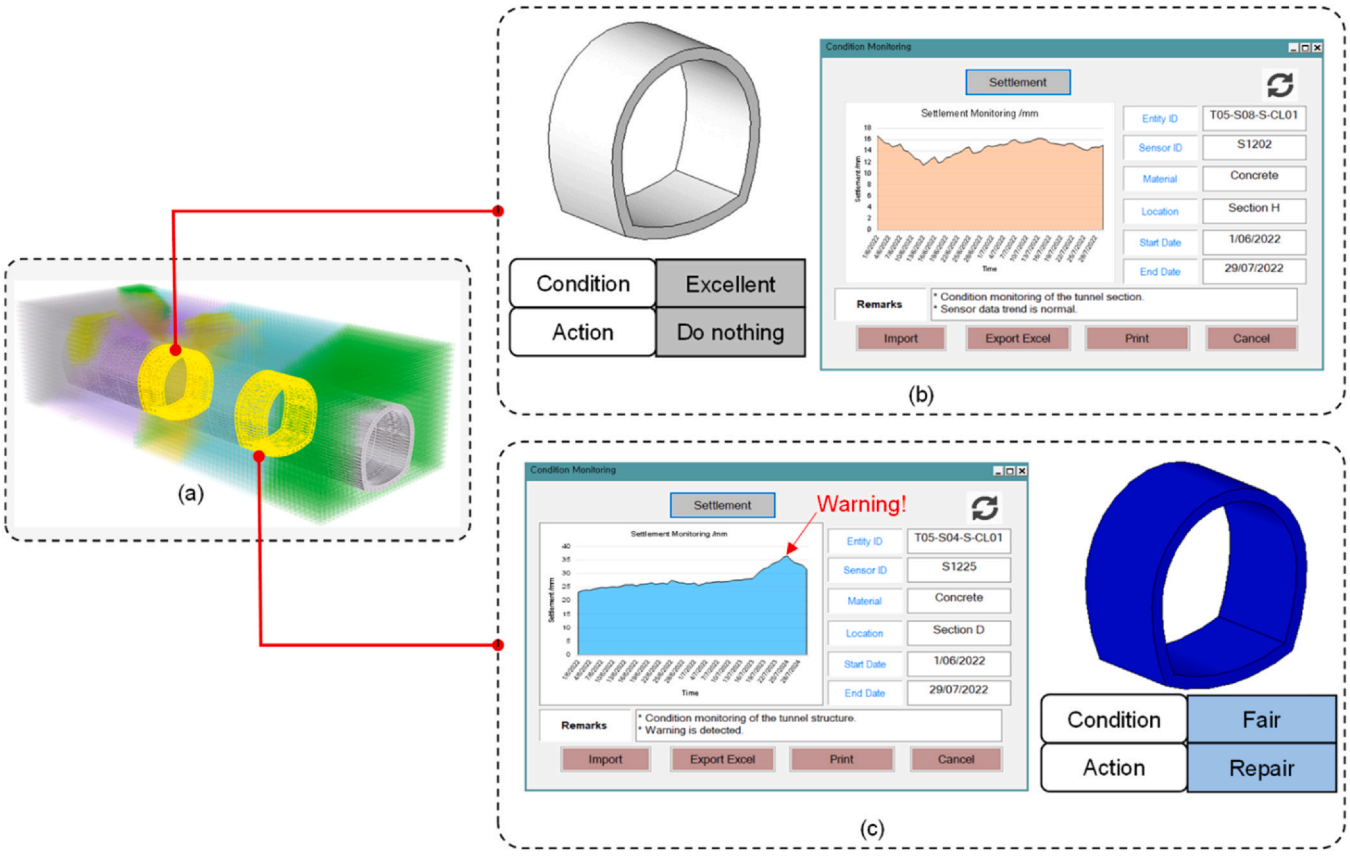


Fig. 14. Condition monitoring of concrete sections (section H and section D), (a) holistic representation of the case study tunnel, (b) settlement variation with condition visualization at section H, and (c) settlement variation with condition visualization at section D.

lines represent the predicted data after applying the SVM and ANN algorithms. Fig. 15a presents the settlement prediction results for Section H, and the settlement results for Section D are shown in Fig. 15b. It also shows the relative error for each measured and predicted value of. A comparison of the measured and predicted values using the SVM and ANN is presented in Tables 3 and 4 for Sections H and D. It can be concluded that the predicted values are close to the measured values, and the accuracies of the SVM and ANN models are 97.621 % and 95.336 %, respectively, for Section H. Consequently, the accuracies of the SVM and ANN models for Section D were 97.755 % and 95.925 %, respectively. This indicated that the predicted values had a high degree of fit with the measured values. For instance, the accuracy of the SVM was better in this case than that of the ANN. Correspondingly, the absolute average error of the SVM was 2.379 %, which is smaller than that of the ANN for Section H (4.664 %).

As shown in Fig. 15a, the fluctuation in the predicted settlement values is small, and the visualization of the model suggests that the section is in a safe state for the overall monitoring period. However, the values are very close to the monitored data for Section D, and minor repair is predicted by the DT system because the settlement value is higher than the threshold set for minor repair (comparison of Figs. 14c and 15b). The maintenance action is predicted at the same time interval as that detected by the measured data (condition monitoring). This validates the accuracy of the proposed system because the predicted warning matches the measured warning. This function helps the FM staff schedule maintenance plans in advance. The small error and high accuracy of the predicted values verify that the proposed system is reliable and can be efficiently used to predict future settlement patterns of tunnel structures.

In addition to accuracy and average error, *RMSE*, *MAE*, and R^2 are also calculated. For section H, the SVM model achieve *RMSE* about

Table 3
Settlement prediction information at Section H.

Data ID	Monitoring date	Measured data	Predicted (SVM)	Error (SVM)	Predicted (ANN)	Error
1	18/7/2022	15.22	15.01	1.37	15.34	0.78
2	19/7/2022	15.09	14.9	1.25	15.05	0.26
3	20/7/2022	15	15.11	0.73	14.67	2.2
4	21/7/2022	15.32	15.56	1.56	15.43	0.71
5	22/7/2022	15.3	15.78	3.13	15.56	1.69
6	23/7/2022	14.89	15.33	2.95	15.36	3.15
7	24/7/2022	14.56	14.25	2.12	14.88	2.19
8	25/7/2022	14.23	14.12	0.77	14.33	0.70
9	26/7/2022	14.11	13.98	0.92	13.86	1.77
10	27/7/2022	14.56	14.66	0.68	14.86	2.06
11	28/7/2022	14.67	14.87	1.36	14.4	1.84
12	29/7/2022	14.65	14.23	2.86	14.76	0.75
13	30/7/2022	15.01	15.2	1.26	14.81	1.33



Fig. 15. Predicting condition of the sections (section H and section D), (a) predicted settlement variation with condition visualization at section H, and (b) predicted settlement variation with condition visualization at section D.

Table 4
Settlement prediction information at Section D.

Data ID	Monitoring time	Measured data	Predicted (SVM)	Error (SVM)	Predicted (ANN)	Error (ANN)
1	18/7/2022	30.76	31.05	0.94	31.33	1.85
2	19/7/2022	31.77	31.34	1.35	30.99	2.45
3	20/7/2022	32.12	31.5	1.93	32.45	1.02
4	21/7/2022	33.45	32.88	1.70	32.99	1.37
5	22/7/2022	33.98	34.4	1.23	34.5	1.53
6	23/7/2022	34.56	33.86	2.03	34.99	1.22
7	24/7/2022	35.98	36.55	1.58	37.1	3.11
8	25/7/2022	36.5	39.09	4.35	36.78	0.76
9	26/7/2022	34.9	35.76	2.46	33.3	4.58
10	27/7/2022	33.9	32.65	3.68	34.43	1.56
11	28/7/2022	33.5	32.45	3.13	34.56	3.16
12	29/7/2022	32.99	33.69	2.12	32.44	1.66
13	30/7/2022	31.54	31.9	1.14	30.44	3.48

0.272 mm, an *MAE* of 0.241 mm, and R^2 of 0.473. On the other hand, *RMSE* for ANN is 0.250 mm, an *MAE* of 0.241 mm, and R^2 of 0.553. For section D, *RMSE* for SVM and ANN were recorded as 0.988 mm and 0.810 mm, an *MAE* of 0.801 mm and 0.718 mm, and R^2 of 0.638 and 0.757. These results highlight several important points. Both SVM and ANN models achieved relatively small error margins to the measurement values recorded using monitoring system. The *RMSE* and *MAE* values for both section H and section D are recorded below one

millimeter. For instance, model performance between two section is different. For section H, SVM perform marginally better in terms of absolute error because of smoother trends and small dataset. In such condition, kernel-based methods often excel. For section D, facing relatively high variability, ANN achieve lower errors and higher R^2, which is consistent with the neural network since they perform better on nonlinear relationships. These variations in the system performance suggests that in a full-scale deployment, the DT system could select the ML according to the as-built condition of the section and the dataset.

The output of the two sets of cases demonstrate the positive performance of the developed DT system in supporting predictive maintenance. The ability of the system to predict maintenance requirements resolves the longstanding issues of service delays and cost overruns caused by traditional maintenance approaches and inefficient FM systems. The proposed DT system is suitable for predicting the maintenance of tunnel components. The performance of prediction algorithms is influenced by various factors, including the quality and quantity of monitored data, as well as the chosen algorithms. In the case study, SVM and ANN algorithms are selected for settlement prediction. Although the accuracy of these algorithms differed, both performed at a sufficiently high level for use in real-world FM practices. In fact, the algorithms successfully predicte warnings within the same timeframe as those detected by the measured data. For instance, this reflects that the chosen features were appropriate for predictive asset management according to the scenerios. Additionally, it demonstrates the reliability of the proposed system for forecasting system failures in advance, enabling FM managers to schedule maintenance and avoid costly downtime.

In addition to accuracy, the other Quantative metrics confirm that the predictive module is capable of supporting proactive maintenance. Other functions of the developed system such as partitioning into section is advantageous here, allowing efficient localized predictions, which are crucial for elongated tunnels. These can significantly contribute in practice for tunnel with varying condition or historical tunnels where maintenance along the length can be prioritized based on the condition. Overall, the system function to integrate multi-source data into each segment, perform the predictions, and integrate it with the 3D model for enhanced visualization can serve as critical tool and foundation for future studies.

6. Conclusions, limitations, and future work

This paper presents a novel data-driven DT-based framework for improving tunnel FM processes. The proposed framework comprises six layers: data acquisition, transmission, modeling, integration, analytics, and application. The proposed DT system design incorporates six modules that provide visualization, data analytics, and ML capabilities, allowing the integration of data from the monitoring system, FM system, and BIM models. Moreover, a tunnel data model and classification system are proposed that hierarchically divides the tunnel models, ensuring an efficient connection of data from the physical twin to the DT. The contributions of this study to existing knowledge include developing and implementing data-driven predictive maintenance strategies that improve tunnel maintenance practices by predicting asset future states, avoiding failures, and creating a smart tunnel maintenance system.

The proposed data-driven DT framework was implemented in a tunnel case study at the operational stage, and multiple scenarios were considered to test different functions of the system. The proposed system successfully connected the physical and digital twins and generated maintenance plans and work orders using historical and current condition monitoring data. Furthermore, the proposed method uses 3D visualization technology to suggest the maintenance and repair processes, thereby making the FM decision-making process more effective. The proposed system not only integrates the condition monitoring data from the physical twin but also detects and predicts the twin state based on data-driven analysis. The prediction results closely matched the measured results, and the prediction accuracy of the ML models was sufficiently high for use in real scenarios to make FM plans in advance and prevent asset failures. The proposed framework provides an innovative solution for enhancing tunnel FM practices by incorporating data-driven methods, predictive maintenance, and 3D visualization technology. This offers a promising approach for optimizing FM decision-making and providing comprehensive FM data, thereby improving the efficiency and effectiveness of tunnel FM.

Despite the advantages, the system has several limitations in the current state. First, the data completeness and reliability in actual tunnel projects may vary because of several reasons such as sensor malfunctions, inconsistent historical records, data unavailability. These factors can impact the predictive model development since the data driven systems are dependent on real-time data feeds. Secondly, although the system was validated on a small section; scaling the system to entire tunnel project could impose additional computational challenges. This huge system would require multiple sensors, database, and BIM information, and their update in real-time may require high-performance cloud computing infrastructure and optimized data processing pipelines. A minor constraint of the current system is the potential latency resulting from network connectivity or high-volume data streams; however, these delays were small in our validation testing and are being mitigated through cloud-based optimization. Finally, integrating data from system such as FM database that may include sensitive information such as contractor identities. Also, sensor feeds may include information about asset vulnerability and operational condition. This raises important data security and privacy concerns and limitations (Chhetri et al., 2024; Torre et al., 2016).

This study primarily addresses tunnel facility management, however the suggested framework is conceptually adaptable to various infrastructure assets, including bridges, highways, rail networks, and subterranean utilities. Numerous facilities management issues are prevalent across infrastructure sectors, encompassing the necessity for real-time monitoring, predictive maintenance, and the amalgamation of diverse data sources. In bridge management, a six-layer framework might amalgamate sensor data from strain gauges, accelerometers, and corrosion sensors with BIM-based digital twins for structural health monitoring. The framework for highways might use IoT-enabled pavement condition monitoring and traffic load data to forecast deterioration trends and enhance maintenance scheduling. Adapting the framework to alternative infrastructure types necessitates domain-specific customization of data models and ontologies, analytics and predictive models to account for varying deterioration mechanisms, as well as integration and visualization tools tailored to the geometry and asset hierarchies of different facilities. For cloud readiness under high-frequency sensor loads, future work can be to define realistic load targets (e.g., 10–20k sensor messages per second per tunnel section, including short bursts), set clear latency goals (p95 ≤ 2–3 s from ingestion to dashboard), and use a simple streaming path (gateway to message queue to short-window processing to time-series store to application/API). Auto scaling and tiered storage (hot/warm/cold) can be used to keep costs predictable while sustaining throughput

Subsequent research ought to investigate the creation of distributed digital twin platforms that may oversee various asset categories inside a unified facility management ecosystem, while guaranteeing comprehensive scalability and interoperability via open standards like IFC and CityGML. A potential domain is the application of edge computing and AI at the sensor level to diminish dependence on cloud infrastructure by integrating lightweight analytics directly at the monitoring nodes. Future endeavors may enhance predictive capabilities by incorporating lifecycle sustainability metrics and resilience indicators, creating immersive mobile and AR/VR applications that enable maintenance teams to visualize asset conditions and obtain on-site decision support, and tackling cybersecurity issues through privacy-preserving computation and secure data-sharing frameworks. By following these guidelines, the

proposed DT framework can develop into a multi-domain infrastructure FM platform that facilitates data-driven and predictive decision-making across various civil engineering assets. This scalability not only amplifies the long-term efficacy of the suggested approach but also bolsters wider smart infrastructure management endeavors. To illustrate scalability, the pipeline can be enhanced through: edge pre-processing (filtering, compression) to reduce traffic, dependable, duplicate-safe ingestion with retries; efficient model serving optimized for sub-second responses at a minimum of 1000 requests per second; safety mechanisms (rate limits, priority queues) for essential channels; and fundamental observability and load testing to provide consistent throughput, latency, and cost per million messages. These steps can be articulated as quantifiable objectives in forthcoming implementations.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Declaration of Generative AI and AI-assisted technologies in the writing process

During the preparation of this work the author used CHATGPT in order to improve English grammar for better readability. After using this tool, the author reviewed and edited the content as needed and take full responsibility for the content of the publication.

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